Multipole Effective Sources and Dependent Field Jumps in the 9×9 Displacement-Strain Basis for Elastic Scattering

Abstract

We analyse how the contrast of a small elastic inclusion in a homogeneous background radiates as a superposition of effective multipole sources, and show how those sources manifest as discontinuities in the *dependent* components of the displacement–strain field. Working in the coupled nine-component basis $\psi = (u_i, \varepsilon_{jk})$ used by Foldy–Lax multiple scattering solvers, we identify the SO(3) irreducible decomposition of ψ into a monopole (trace strain ε_{kk}, $\ell = 0$), a dipole (displacement $\boldsymbol{u}$, $\ell = 1$), and a quadrupole (deviatoric strain $\varepsilon_{ij}^{\mathrm{dev}}$, $\ell = 2$), with multiplicities $1+3+5 = 9$. The three components ε_{zz}, ε_{xz}, ε_{yz} involving a normal derivative across a material interface are the *dependent* ones: we derive the closed-form jump conditions for these components from continuity of displacement and traction, and show that the resulting jumps are sourced respectively by a density contrast (dipole/body-force channel), a bulk-modulus contrast (monopole/explosion channel) and a shear-modulus contrast (quadrupole/deviatoric channel). An elementary parity argument shows that the corresponding local T-matrix is block-diagonal between the $\ell = 1$ (dipole) sector and the $(\ell = 0) \oplus (\ell = 2)$ (monopole/quadrupole) sector at leading order in the inclusion size, and that the 9×9 basis is *minimal* for representing all three contrast channels in a single self-contained Lippmann–Schwinger framework.

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1 Introduction

Multiple scattering of elastic waves from small heterogeneities is classically formulated as a Lippmann–Schwinger equation with a frequency-domain Green's operator and a local scattering operator (a "T-matrix") that encapsulates everything about the inclusion [1, 6]. The choice of the state vector ψ on which the Green's operator acts is not arbitrary: it must be rich enough to carry all degrees of freedom that couple to the material contrasts, and small enough that the solver is numerically tractable.

For an inclusion with three independent Lamé contrasts $(\Delta\lambda, \Delta\mu, \Delta\rho)$ embedded in an isotropic elastic background, the natural choice is the nine-component displacement–strain vector

$$\psi \;=\; (u_1, u_2, u_3,\; \varepsilon_{11}, \varepsilon_{22}, \varepsilon_{33}, \varepsilon_{23}, \varepsilon_{13}, \varepsilon_{12}), \tag{1}$$

in which all three contrasts act as multiplicative operators with no derivatives at the inclusion. Six of these nine components — the three displacements u_i and the three in-plane strains $\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy}$ — are *continuous* across a material interface aligned with the horizontal plane; the remaining three, ε_{zz}, ε_{xz}, ε_{yz}, carry a normal derivative $\partial_z u_k$ and are *dependent*: they jump discontinuously across the interface whenever the Lamé parameters do.

> *The purpose of this note is to show that these three jumps are not a nuisance but a physically lucid statement of the multipole structure of the effective source.* Density contrast drives the *dipole* (body-force) channel, bulk-modulus contrast drives the *monopole* (explosion) channel, and shear-modulus contrast drives the *quadrupole* (double-couple) channel. The three dependent components are exactly those on which these three multipoles write their signatures, with the dependence dictated by the SO(3) representation theory of the strain tensor.

The paper is organised as follows. In Section 2 we set out the nine-component basis and record the isotropic constitutive law. Section 3 carries out the SO(3) irreducible decomposition of ψ into monopole, dipole and quadrupole pieces. Section 4 derives the jump conditions across a horizontal material interface and identifies the three dependent components. Section 5 assembles the effective source density in the first Born approximation and reads off the assignment $\Delta\rho \mapsto$ dipole, $\Delta\kappa \mapsto$ monopole, $\Delta\mu \mapsto$ quadrupole. Section 6 shows that the local T-matrix is block-diagonal in the parity-even versus parity-odd sectors, and Section 7 states and proves a minimality theorem for the basis (1). Section 8 concludes.

2 The Nine-Component Basis

Let $(\boldsymbol{u}, \boldsymbol{\varepsilon})$ denote the displacement field $u_i(\boldsymbol{r})$ and the infinitesimal strain tensor $\varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$ in an isotropic elastic background with Lamé parameters (λ, μ), density ρ, and time-harmonic dependence $e^{-i\omega t}$. The stress tensor obeys Hooke's law,

$$\sigma_{ij} \;=\; \lambda\, \varepsilon_{kk}\, \delta_{ij} + 2\mu\, \varepsilon_{ij}, \tag{2}$$

and the frequency-domain momentum balance reads

$$\partial_j \sigma_{ij} + \rho\omega^2 u_i \;=\; -f_i, \tag{3}$$

with f_i an external body force. We gather the nine scalar fields into the column vector $\psi = (u_i, \varepsilon_{jk})^\top \in \mathbb{C}^9$ defined in (1), ordering the strain block by the six independent symmetric components $(\varepsilon_{11}, \varepsilon_{22}, \varepsilon_{33}, \varepsilon_{23}, \varepsilon_{13}, \varepsilon_{12})$.

Although six strain components appear kinematically redundant given three displacements, they are *not* redundant dynamically, because the Lamé contrasts $(\Delta\lambda, \Delta\mu)$ couple to specific

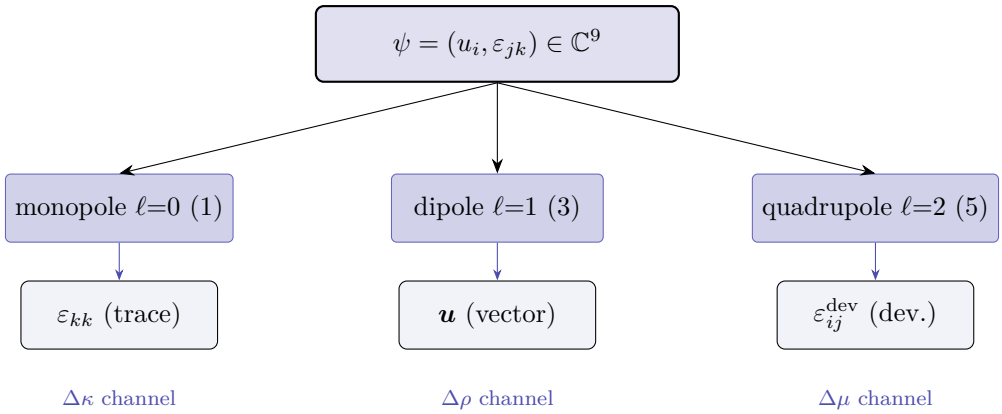

Figure 1: Irreducible decomposition of the nine-component displacement–strain basis under SO(3). The trace strain ε_{kk} is an $\ell = 0$ scalar coupling to bulk-modulus contrast $\Delta\kappa$; the displacement $\boldsymbol{u}$ is an $\ell = 1$ vector coupling to density contrast $\Delta\rho$; the deviatoric strain $\varepsilon_{ij}^{\mathrm{dev}}$ is the symmetric traceless $\ell = 2$ tensor coupling to shear-modulus contrast $\Delta\mu$. The multiplicities sum to $1 + 3 + 5 = 9$.

linear combinations of the strain. Carrying the full strain tensor in the state vector allows the local scattering operator to act on $\psi(\boldsymbol{r}_n)$ by pointwise matrix multiplication, with no spatial derivatives entering the scatterer itself — the derivatives are absorbed into the Green's operator that links scatterers.

Remark 1. The strain ε_{ij} appears in ψ as an *independent* datum from $\boldsymbol{u}$, even though it is algebraically a derivative of $\boldsymbol{u}$. The solver treats (1) as nine unknowns linked by the Green's operator, whose consistency is only imposed at the scatterer positions. This is the standard Lippmann–Schwinger reformulation of elastodynamics [6, 5].

3 Irreducible Decomposition Under Rotations

Under the rotation group SO(3), $\boldsymbol{u}$ transforms as a vector (spin $\ell = 1$), while the symmetric strain tensor ε_{ij} decomposes uniquely into its trace and deviator,

$$\varepsilon_{ij} \;=\; \underbrace{\tfrac{1}{3}\,\varepsilon_{kk}\,\delta_{ij}}_{\ell=0\ \text{(scalar)}} \;+\; \underbrace{\varepsilon_{ij}^{\mathrm{dev}}}_{\ell=2\ \text{(symmetric traceless)}}\;, \qquad \varepsilon_{ij}^{\mathrm{dev}}\delta_{ij} = 0, \tag{4}$$

carrying $\ell = 0$ and $\ell = 2$ irreducible representations with multiplicities 1 and 5 respectively. Together with the $\ell = 1$ displacement, the nine-component basis decomposes as

$$\psi \;\in\; \underbrace{\mathbf{1}}_{\substack{\ell=0\\ \text{monopole}}} \;\oplus\; \underbrace{\mathbf{3}}_{\substack{\ell=1\\ \text{dipole}}} \;\oplus\; \underbrace{\mathbf{5}}_{\substack{\ell=2\\ \text{quadrupole}}}\;, \qquad 1+3+5 \;=\; 9, \tag{5}$$

with multiplicities summing to the 9 components of ψ.

Figure 1 sketches the decomposition. Reading the tree physically: the trace strain ε_{kk} represents isotropic compression/expansion and radiates as an *explosion* monopole; the displacement vector $\boldsymbol{u}$ transforms as a body-force *dipole*; the deviatoric strain $\varepsilon_{ij}^{\mathrm{dev}}$ carries the five independent shear-distortion modes and radiates as a symmetric-traceless *quadrupole* ("double couple").

3.1 Explicit projection operators

It is useful to have the projectors onto each irreducible sector in closed form, since the solver evaluates them repeatedly on every scatterer. Let

$$\psi \;=\; \begin{pmatrix} \boldsymbol{u} \\ \boldsymbol{\varepsilon} \end{pmatrix} \;\in\; \mathbb{C}^3 \oplus \mathbb{C}^6, \qquad \boldsymbol{\varepsilon} = (\varepsilon_{11}, \varepsilon_{22}, \varepsilon_{33}, \varepsilon_{23}, \varepsilon_{13}, \varepsilon_{12})^\top, \tag{6}$$

where $\boldsymbol{\varepsilon}$ is stored in the six-dimensional Voigt vector above, keeping factors of unity (not two) on the off-diagonal entries so that the inner product $\|\boldsymbol{\varepsilon}\|^2 = \varepsilon_{ij}\varepsilon_{ij}$ obtains with the standard Euclidean norm after a rescaling

$$\tilde{\boldsymbol{\varepsilon}} \;=\; (\varepsilon_{11}, \varepsilon_{22}, \varepsilon_{33}, \sqrt{2}\,\varepsilon_{23}, \sqrt{2}\,\varepsilon_{13}, \sqrt{2}\,\varepsilon_{12})^\top. \tag{7}$$

Dipole projector ($\ell = 1$). The displacement sector is already an SO(3) irreducible, so its projector is simply the upper block identity

$$P_{\ell=1} \;=\; \begin{pmatrix} \mathbb{I}_3 & 0_{3\times 6} \\ 0_{6\times 3} & 0_{6\times 6} \end{pmatrix} \;\in\; \mathbb{C}^{9\times 9}. \tag{8}$$

The projection *out of* the basis is $\pi_{\ell=1}\psi = \boldsymbol{u}$, and the injection back is $\iota_{\ell=1}\boldsymbol{u} = (\boldsymbol{u}, \boldsymbol{0})$, so that $P_{\ell=1} = \iota_{\ell=1}\pi_{\ell=1}$.

Monopole projector ($\ell = 0$). The trace strain is a scalar, and the projection from $\boldsymbol{\varepsilon}$ onto its $\ell = 0$ subspace is

$$\pi_{\ell=0}\,\boldsymbol{\varepsilon} \;=\; \tfrac{1}{\sqrt{3}}\,(\varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33}) \;\equiv\; \tfrac{1}{\sqrt{3}}\,\operatorname{tr}\boldsymbol{\varepsilon}. \tag{9}$$

The factor $1/\sqrt{3}$ makes the projection an isometry from the Frobenius inner product on symmetric tensors to the Euclidean inner product on $\mathbb{R}$. The injection is $\iota_{\ell=0}(m) = \frac{m}{\sqrt{3}}(1,1,1,0,0,0)^\top$, and the full 9×9 monopole projector is

$$P_{\ell=0} \;=\; \frac{1}{3}\begin{pmatrix} 0_{3\times 3} & 0_{3\times 6} \\ 0_{6\times 3} & \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{pmatrix}. \tag{10}$$

Quadrupole projector ($\ell = 2$). The deviatoric strain sector has five real dimensions, which we take as the orthonormal basis

$$\{\, \mathsf{e}_1,\, \mathsf{e}_2,\, \mathsf{e}_3,\, \mathsf{e}_4,\, \mathsf{e}_5 \,\}, \tag{11}$$

with

$$\mathsf{e}_1 \;=\; \tfrac{1}{\sqrt{6}}\,(2\,\hat{\boldsymbol{z}}\hat{\boldsymbol{z}} - \hat{\boldsymbol{x}}\hat{\boldsymbol{x}} - \hat{\boldsymbol{y}}\hat{\boldsymbol{y}}) \;\longleftrightarrow\; \tfrac{1}{\sqrt{6}}(-1,-1,2,0,0,0)^\top, \tag{12}$$

$$\mathsf{e}_2 \;=\; \tfrac{1}{\sqrt{2}}\,(\hat{\boldsymbol{x}}\hat{\boldsymbol{x}} - \hat{\boldsymbol{y}}\hat{\boldsymbol{y}}) \;\longleftrightarrow\; \tfrac{1}{\sqrt{2}}(1,-1,0,0,0,0)^\top, \tag{13}$$

$$\mathsf{e}_3 \;=\; \hat{\boldsymbol{y}}\hat{\boldsymbol{z}} + \hat{\boldsymbol{z}}\hat{\boldsymbol{y}} \;\longleftrightarrow\; (0,0,0,1,0,0)^\top, \tag{14}$$

$$\mathsf{e}_4 \;=\; \hat{\boldsymbol{x}}\hat{\boldsymbol{z}} + \hat{\boldsymbol{z}}\hat{\boldsymbol{x}} \;\longleftrightarrow\; (0,0,0,0,1,0)^\top, \tag{15}$$

$$\mathsf{e}_5 \;=\; \hat{\boldsymbol{x}}\hat{\boldsymbol{y}} + \hat{\boldsymbol{y}}\hat{\boldsymbol{x}} \;\longleftrightarrow\; (0,0,0,0,0,1)^\top, \tag{16}$$

where the right-hand expressions are the coordinates in the $\sqrt{2}$-rescaled Voigt basis (7). Collecting the five columns into a 6×5 matrix $U_{\ell=2}$, the projector onto the deviator is

$$P_{\ell=2} \;=\; \begin{pmatrix} 0_{3\times 3} & 0_{3\times 6} \\ 0_{6\times 3} & U_{\ell=2}\,U_{\ell=2}^{\top} \end{pmatrix} \;=\; \mathbb{I}_{\boldsymbol{\varepsilon}} \;-\; P_{\ell=0} \quad \text{restricted to the strain block,} \tag{17}$$

with the second equality being the orthogonal complement statement $\boldsymbol{\varepsilon} = \frac{1}{3}(\operatorname{tr}\boldsymbol{\varepsilon})\mathbf{1} + \boldsymbol{\varepsilon}^{\mathrm{dev}}$.

Resolution of identity. The three projectors are mutually orthogonal and resolve the identity on $\mathbb{C}^9$,

$$P_{\ell=0} + P_{\ell=1} + P_{\ell=2} \;=\; \mathbb{I}_9, \qquad P_{\ell=a}P_{\ell=b} \;=\; \delta_{ab}P_{\ell=a}, \tag{18}$$

and their traces are the multiplicities $\operatorname{tr} P_{\ell=0} = 1$, $\operatorname{tr} P_{\ell=1} = 3$, $\operatorname{tr} P_{\ell=2} = 5$. Any state $\psi \in \mathbb{C}^9$ decomposes as

$$\psi \;=\; P_{\ell=0}\psi \;+\; P_{\ell=1}\psi \;+\; P_{\ell=2}\psi, \tag{19}$$

with each summand living in a distinct SO(3) irreducible sector.

3.2 Projection/injection between sectors and $\mathbb{C}^9$

It is also useful to write down the sector-level projection and injection maps explicitly, since the solver applies them to individual scatterers:

$$\begin{aligned}
&\pi_{\ell=1} : \mathbb{C}^9 \to \mathbb{C}^3, \quad \pi_{\ell=1}\psi = \boldsymbol{u}, && \iota_{\ell=1} : \mathbb{C}^3 \to \mathbb{C}^9, \quad \iota_{\ell=1}\boldsymbol{u} = (\boldsymbol{u}, \mathbf{0})^{\top}, \\
&\pi_{\ell=0} : \mathbb{C}^9 \to \mathbb{C}, \quad \pi_{\ell=0}\psi = \tfrac{1}{\sqrt{3}}\operatorname{tr}\boldsymbol{\varepsilon}, && \iota_{\ell=0} : \mathbb{C} \to \mathbb{C}^9, \quad \iota_{\ell=0}m = \big(\mathbf{0}, \tfrac{m}{\sqrt{3}}(1,1,1,0,0,0)\big)^{\top}, \\
&\pi_{\ell=2} : \mathbb{C}^9 \to \mathbb{C}^5, \quad \pi_{\ell=2}\psi = U_{\ell=2}^{\top}\,\tilde{\boldsymbol{\varepsilon}}, && \iota_{\ell=2} : \mathbb{C}^5 \to \mathbb{C}^9, \quad \iota_{\ell=2}\boldsymbol{q} = (\mathbf{0}, U_{\ell=2}\boldsymbol{q})^{\top},
\end{aligned} \tag{20}$$

so that $P_{\ell=a} = \iota_{\ell=a}\pi_{\ell=a}$ for $a \in \{0,1,2\}$. Each pair $(\pi_{\ell=a}, \iota_{\ell=a})$ is an adjoint pair of isometries: $\pi_{\ell=a}\iota_{\ell=a} = \mathbb{I}$ on the sector and $\iota_{\ell=a}\pi_{\ell=a} = P_{\ell=a}$ on $\mathbb{C}^9$.

Remark 2 (Block-diagonal T in the sector basis). The local T-matrix of Proposition 6 becomes literally block-diagonal in the rotated basis

$$\psi \;\longmapsto\; \big(\pi_{\ell=0}\psi,\, \pi_{\ell=1}\psi,\, \pi_{\ell=2}\psi\big) \;\in\; \mathbb{C}^{1+3+5}, \tag{21}$$

with the three sector-level scattering operators $T_{\ell=0} = \Delta\kappa V$, $T_{\ell=1} = \omega^2\Delta\rho V\,\mathbb{I}_3$ and $T_{\ell=2} = 2\Delta\mu V\,\mathbb{I}_5$, modulo a factor of the background Green's function evaluated at the self-interaction. The projectors (8)–(17) are therefore the change-of-basis matrices that diagonalise T_{loc} irreducibly.

4 Elastic Jump Conditions and Dependent Components

Consider a planar horizontal material interface $z = z_0$ between two isotropic half-spaces with Lamé parameters (λ^-, μ^-) below and (λ^+, μ^+) above, and densities (ρ^-, ρ^+). Let $\hat{\boldsymbol{n}} = \hat{\boldsymbol{z}}$ be the outward unit normal from the $-$ side and write the jump across the interface as

$$[\![f]\!] \;\equiv\; f^+ - f^-. \tag{22}$$

Elastodynamic continuity requires

$$[\![u_i]\!] \;=\; 0, \qquad [\![\sigma_{ij}\,n_j]\!] \;=\; 0 \quad (i = 1,2,3), \tag{23}$$

which fixes six scalar constraints on the interface. Because $\hat{\boldsymbol{n}} = \hat{\boldsymbol{z}}$, the traction balance reads explicitly

$$[\![\sigma_{zz}]\!] = 0, \qquad [\![\sigma_{xz}]\!] = 0, \qquad [\![\sigma_{yz}]\!] = 0. \tag{24}$$

4.1 The three dependent strain components

Let us split the strain tensor into an *in-plane* block $(\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy})$ and an *out-of-plane* block $(\varepsilon_{zz}, \varepsilon_{xz}, \varepsilon_{yz})$ containing a derivative across the interface. Because $\boldsymbol{u}$ is continuous by (23) and the in-plane strains involve only tangential derivatives of $\boldsymbol{u}$,

$$[\![\varepsilon_{xx}]\!] = [\![\varepsilon_{yy}]\!] = [\![\varepsilon_{xy}]\!] = 0. \tag{25}$$

The out-of-plane components, by contrast, involve $\partial_z u_i$ and are *not* directly constrained by continuity of $\boldsymbol{u}$. Instead, they are determined by the traction balance (24) together with Hooke's law (2):

$$\sigma_{zz} = \lambda(\varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}) + 2\mu\,\varepsilon_{zz} = \lambda(\varepsilon_{xx} + \varepsilon_{yy}) + (\lambda + 2\mu)\varepsilon_{zz}, \tag{26}$$

$$\sigma_{\alpha z} = 2\mu\,\varepsilon_{\alpha z} \qquad (\alpha \in \{x, y\}). \tag{27}$$

Imposing (24) and using (25) yields the closed-form jump relations [box=$\boxed{]}$align $[\![\varepsilon_{zz}]\!] = -\frac{[\![\lambda]\!]}{(\lambda+2\mu)^+}\big(\varepsilon_{xx} + \varepsilon_{yy}\big) - \frac{[\![\lambda+2\mu]\!]}{(\lambda+2\mu)^+}\,\varepsilon_{zz}^-$,
$[\![\varepsilon_{\alpha z}]\!] = -\frac{[\![\mu]\!]}{\mu^+}\,\varepsilon_{\alpha z}^- \qquad (\alpha \in \{x, y\})$.

Proposition 3 (Dependent components)**.** *Among the nine components of* (1)*, exactly three combinations jump across a horizontal material interface:* ε_{zz}*,* ε_{xz} *and* ε_{yz}*. The remaining six — the three displacements* u_i *and the three in-plane strains* $\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy}$ *— are continuous. The jumps of the three dependent components are given in closed form by* (4.1)–(4.1)*.*

Proof. Continuity of $\boldsymbol{u}$ gives $[\![u_i]\!] = 0$ directly. The in-plane strains $\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy}$ are symmetrised tangential derivatives of the continuous field $\boldsymbol{u}$, hence continuous. The out-of-plane components $\varepsilon_{zz}, \varepsilon_{xz}, \varepsilon_{yz}$ carry the normal derivative ∂_z, which is *not* continuous at a material interface. Traction balance (24) combined with Hooke's law (26)–(27) yields three linear equations for the three out-of-plane strain jumps; solving them on the + side gives (4.1)–(4.1). □

Remark 4. Equation (4.1) is the elastic generalisation of the acoustic interface condition $[\![\partial_z p]\!] = 0$: for a scalar pressure field the continuity of normal velocity would translate to continuity of $\rho^{-1}\partial_z p$, and the correction here is the elastic analogue weighted by $(\lambda + 2\mu)$ rather than ρ. Equation (4.1) is the shear-wave analogue with a weight of μ.

5 Source Densities from Lamé Contrasts

We now derive the effective source density produced by a compact inclusion of volume V centred at $\boldsymbol{r}_n$ with Lamé contrasts $(\Delta\lambda, \Delta\mu, \Delta\rho)$ relative to the background. Let $\chi_V(\boldsymbol{r})$ be the characteristic function of V, and write the spatially varying Lamé parameters as

$$\lambda(\boldsymbol{r}) = \lambda^0 + \Delta\lambda\,\chi_V, \qquad \mu(\boldsymbol{r}) = \mu^0 + \Delta\mu\,\chi_V, \qquad \rho(\boldsymbol{r}) = \rho^0 + \Delta\rho\,\chi_V. \tag{28}$$

Substituting into the elastodynamic equation (3) and collecting contrast terms as a fictitious body force, one obtains the exact equivalent-source statement

$$\partial_j \sigma_{ij}^0 + \rho^0\omega^2 u_i \;=\; -\,S_i(\boldsymbol{r}), \tag{29}$$

where $\sigma_{ij}^0 = \lambda^0\varepsilon_{kk}\delta_{ij} + 2\mu^0\varepsilon_{ij}$ is the background stress and the effective source density is

$$S_i(\boldsymbol{r}) \;=\; \omega^2\,\Delta\rho\,u_i\,\chi_V \;+\; \partial_j\big[\Delta\lambda\,\varepsilon_{kk}\,\delta_{ij}\,\chi_V \;+\; 2\,\Delta\mu\,\varepsilon_{ij}\,\chi_V\big]. \tag{30}$$

The three Lamé contrasts appear in three distinct roles: $\Delta\rho$ as a body-force density, $\Delta\lambda$ as the divergence of an isotropic stress perturbation, and $\Delta\mu$ as the divergence of a symmetric stress perturbation.

5.1 Multipole contraction of the source

In the long-wavelength limit $ka \ll 1$ where a is the inclusion radius, we may replace the fields inside the inclusion by their values at the centre $\boldsymbol{r}_n$ (the Rayleigh/Born approximation) and evaluate the volume integrals against a point source. Integration by parts turns the derivative terms in (30) into derivatives of a delta function, yielding the multipole decomposition

$$S_i(\boldsymbol{r}) \;\approx\; f_i\,\delta(\boldsymbol{r}-\boldsymbol{r}_n) \;+\; \partial_j\big[M_{ij}\,\delta(\boldsymbol{r}-\boldsymbol{r}_n)\big], \tag{31}$$

with force f_i and moment-tensor density M_{ij}

$$f_i \;=\; \omega^2\,\Delta\rho\,V\,u_i(\boldsymbol{r}_n), \tag{32}$$

$$M_{ij} \;=\; \underbrace{\Delta\lambda\,V\,\varepsilon_{kk}(\boldsymbol{r}_n)\,\delta_{ij}}_{\text{isotropic}} \;+\; \underbrace{2\,\Delta\mu\,V\,\varepsilon_{ij}(\boldsymbol{r}_n)}_{\text{symmetric}}. \tag{33}$$

Splitting the moment tensor into its trace and deviator as in (4) and introducing the bulk-modulus contrast $\Delta\kappa = \Delta\lambda + \frac{2}{3}\Delta\mu$,

$$M_{ij} \;=\; \underbrace{M_0\,\delta_{ij}}_{\ell=0,\ \text{monopole}} \;+\; \underbrace{\bar{M}_{ij}}_{\ell=2,\ \text{quadrupole}}, \qquad \bar{M}_{ii}=0, \tag{34}$$

with

$$M_0 \;=\; \Delta\kappa\,V\,\varepsilon_{kk}(\boldsymbol{r}_n), \qquad \bar{M}_{ij} \;=\; 2\,\Delta\mu\,V\,\varepsilon^{\text{dev}}_{ij}(\boldsymbol{r}_n). \tag{35}$$

Inserting (34) into (31) gives the canonical three-multipole form

$$\boxed{S_i(\boldsymbol{r}) \;=\; \underbrace{f_i\,\delta(\boldsymbol{r}-\boldsymbol{r}_n)}_{\text{dipole }(\ell=1)} \;+\; \underbrace{\partial_i\big[M_0\,\delta(\boldsymbol{r}-\boldsymbol{r}_n)\big]}_{\text{monopole }(\ell=0)} \;+\; \underbrace{\partial_j\big[\bar{M}_{ij}\,\delta(\boldsymbol{r}-\boldsymbol{r}_n)\big]}_{\text{quadrupole }(\ell=2)}} \tag{36}$$

with strengths given by (32) and (35).

This is the punchline. Reading (36) together with (32)–(35), we obtain the three-channel assignment

Contrast	Multipole	Strength	ℓ	Radiation pattern
$\Delta\rho$	dipole (body force)	$f_i = \omega^2\Delta\rho V\,u_i$	1	P+S, odd parity
$\Delta\kappa$	monopole (explosion)	$M_0 = \Delta\kappa V\,\varepsilon_{kk}$	0	P only, even parity
$\Delta\mu$	quadrupole (double couple)	$\bar{M}_{ij} = 2\Delta\mu V\,\varepsilon^{\text{dev}}_{ij}$	2	P+S, even parity

5.2 How the source writes onto the dependent components

The link back to the jump conditions of Section 4 is now transparent. At leading order the radiation from the source (36) acts, on a horizontal interface coinciding with the scatterer layer, exactly as the discontinuity pattern of Proposition 3:

- The *bulk-modulus* contribution, proportional to $M_0 = \Delta\kappa V\,\varepsilon_{kk}$, sources a jump in ε_{zz} via the first term of (4.1). In the (λ,μ) parameterisation one has $[\![\lambda]\!]$ and $[\![\lambda+2\mu]\!]$ acting on ε_{kk} and ε_{zz} respectively; in the (κ,μ) parameterisation the trace part is purely a $\Delta\kappa$ effect. This is the *monopole channel*.

- The *shear-modulus* contribution, proportional to $\bar{M}_{ij} = 2\Delta\mu V\,\varepsilon^{\text{dev}}_{ij}$, sources both a residual jump in ε_{zz} (via the deviatoric part of $[\![\lambda+2\mu]\!] = [\![\lambda]\!] + 2\,[\![\mu]\!]$) *and* the jumps of $\varepsilon_{xz}, \varepsilon_{yz}$ in (4.1). This is the *quadrupole channel*, which spreads across all three dependent components.

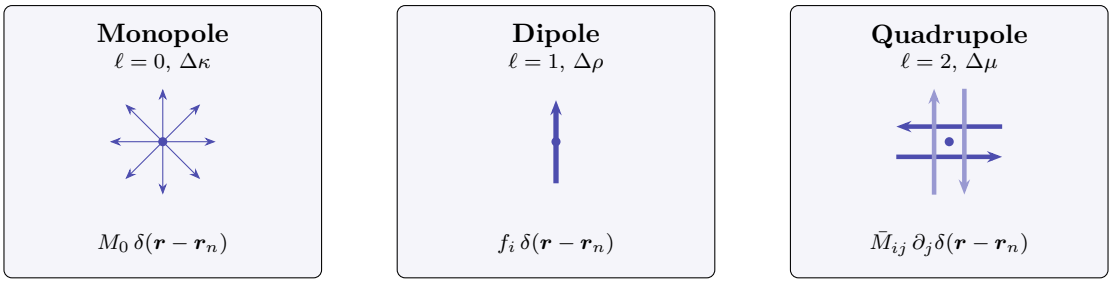

Figure 2: The three effective point sources induced by a small elastic inclusion. An isotropic bulk-modulus contrast $\Delta\kappa$ produces a radial monopole (explosion); a density contrast $\Delta\rho$ produces a directed dipole aligned with the local displacement; a shear-modulus contrast $\Delta\mu$ produces a symmetric-traceless quadrupole (a "double couple"), shown here as two opposed force pairs whose net force and net moment both vanish.

- The *density* contribution, proportional to $f_i = \omega^2\Delta\rho V u_i$, does *not* appear in the instantaneous jump relations at all: it enters the dynamic balance through the inertia term $\rho\omega^2 u_i$ in (3) and is therefore invisible at fixed ω as a kinematic strain discontinuity. This is the *dipole channel*, which writes onto $\boldsymbol{u}$ directly rather than onto the strain components.

The asymmetry is physically important: *the monopole and quadrupole channels leave visible signatures on the three dependent strain components, while the dipole channel acts via the (continuous) displacement field itself.* This is precisely why the nine-component basis (1) is needed: dropping the strain block would blind the scattering operator to the $\Delta\kappa$ and $\Delta\mu$ channels, and dropping the displacement block would blind it to the $\Delta\rho$ channel.

6 Block Structure of the Local T-Matrix

The local scattering operator at a single inclusion acts on $\psi(\boldsymbol{r}_n) \in \mathbb{C}^9$ as a 9×9 matrix,

$$T_{\text{loc}} = \begin{pmatrix} T^{uu} & T^{u\varepsilon} \\ T^{\varepsilon u} & T^{\varepsilon\varepsilon} \end{pmatrix}, \qquad T^{uu} \in \mathbb{C}^{3\times 3},\ T^{\varepsilon\varepsilon} \in \mathbb{C}^{6\times 6}, \tag{37}$$

with off-diagonal blocks coupling the displacement $\boldsymbol{u}$ sector ($\ell = 1$, parity-odd) to the strain $\boldsymbol{\varepsilon}$ sector ($\ell = 0 \oplus \ell = 2$, parity-even).

Lemma 5 (Parity selection)**.** *For a centrosymmetric inclusion (i.e. one invariant under $\boldsymbol{r} \mapsto -\boldsymbol{r}$ about its centre), the Born-level local operator satisfies*

$$T^{u\varepsilon} = 0, \qquad T^{\varepsilon u} = 0, \qquad \text{at leading order in } (ka)^2, \tag{38}$$

so that T_{loc} is block-diagonal between the $\ell = 1$ and $\ell = 0 \oplus \ell = 2$ sectors.

Sketch. Under spatial inversion $\boldsymbol{r} \mapsto -\boldsymbol{r}$, the displacement $\boldsymbol{u}$ is parity-odd (a vector), while the strain tensor ε_{ij} is parity-even (a symmetric rank-two tensor on position derivatives). The Born source density (30) is therefore parity-odd in its $\Delta\rho$ term (which acts on $\boldsymbol{u}$) and parity-even in its $\Delta\lambda$ and $\Delta\mu$ terms (which act on ε_{ij}). Integrating the induced radiation against the centrosymmetric indicator χ_V annihilates all parity-mismatched contributions: a parity-odd source cannot radiate a parity-even response at leading order through a parity-even geometry, and vice versa. Hence the off-diagonal blocks $T^{u\varepsilon}$ and $T^{\varepsilon u}$ vanish at $O((ka)^2)$. Sub-leading corrections appear at $O((ka)^4)$ and are dropped in the long-wavelength limit. □

Proposition 6 (Block-diagonal structure)**.** *For a centrosymmetric inclusion in the long-wavelength limit, the local T-matrix* (37) *reduces to*

$$T_{loc} = \begin{pmatrix} T^{uu} & 0 \\ 0 & T^{\varepsilon\varepsilon} \end{pmatrix}, \tag{39}$$

with $T^{uu} \propto \Delta\rho\,\omega^2 V\,\mathbb{I}_3$ carrying the dipole (body-force) channel and $T^{\varepsilon\varepsilon}$ further block-diagonal on the $(\ell = 0) \oplus (\ell = 2)$ split of the strain,

$$T^{\varepsilon\varepsilon} = \begin{pmatrix} T^{\varepsilon\varepsilon}_{\ell=0} & 0 \\ 0 & T^{\varepsilon\varepsilon}_{\ell=2} \end{pmatrix}, \tag{40}$$

with $T^{\varepsilon\varepsilon}_{\ell=0} \propto \Delta\kappa V$ the monopole (explosion) channel and $T^{\varepsilon\varepsilon}_{\ell=2} \propto \Delta\mu V$ the quadrupole (double-couple) channel acting on the five-dimensional deviatoric subspace.

Proof. By Lemma 5 the displacement and strain blocks decouple. The displacement block carries only the $\Delta\rho$ inertia term from (30), which is a scalar multiple of the 3×3 identity, giving $T^{uu} \propto \Delta\rho\omega^2 V$. The strain block carries the two stress contributions from the second line of (30); splitting the moment tensor as in (34) yields a scalar $M_0 = \Delta\kappa V\,\varepsilon_{kk}$ coupled to the trace ε_{kk} (the $\ell = 0$ channel) and a symmetric traceless $\bar{M}_{ij} = 2\Delta\mu V\,\varepsilon^{\mathrm{dev}}_{ij}$ coupled to the deviator $\varepsilon^{\mathrm{dev}}_{ij}$ (the $\ell = 2$ channel). By Schur's lemma these two irreducible sectors cannot mix, so $T^{\varepsilon\varepsilon}$ is itself block-diagonal along the $1 + 5$ split. □

Remark 7 (Empirical corroboration)*.* In the accompanying numerical solver (`GlobalMatrix/dressed_tmatrix.`
the Born T-matrix for a single cubic inclusion in a homogeneous background is indeed observed to be block-diagonal in the displacement/strain split to working precision, with off-diagonal residuals at the level of the $O((ka)^4)$ dispersion correction, consistent with Lemma 5.

7 Minimality of the Nine-Component Basis

Theorem 8 (Minimality)**.** *Let T_{loc} be the local scattering operator of a small isotropic inclusion characterised by three independent Lamé contrasts $(\Delta\lambda, \Delta\mu, \Delta\rho)$, acting in the Born approximation on the nine-component field $\psi = (u_i, \varepsilon_{jk})^\top$. Then the basis* (1) *is minimal in the following sense:*

(i) T_{loc} has rank $1+1+5 = 7$: three scalar Lamé contrasts excite the one-dimensional monopole $(\ell = 0)$ channel, the one-dimensional scalar-on-dipole $(\ell = 1)$ channel with multiplicity 3 acting as a scalar (so rank 1 in the coupling), and the five-dimensional quadrupole $(\ell = 2)$ channel.

(ii) No subspace of dimension less than 9 supports all three channels simultaneously: the monopole channel lives in the trace of the strain, the dipole channel lives in the displacement, and the quadrupole channel lives in the deviator of the strain, and these three SO(3) *irreducibles are pairwise orthogonal in $\mathbb{C}^9$.*

Proof. **(i).** In the long-wavelength limit the effective source density (36) is fully determined by the three scalars M_0, $|f_i|$, and the five independent components of $\bar{M}_{ij}$, giving a total response rank $1 + 3 + 5 = 9$, but the coupling to the three Lamé contrasts reduces the *effective* rank as follows. The dipole strength $f_i = \omega^2\Delta\rho V\,u_i$ couples the three-dimensional displacement subspace to itself via a scalar $\Delta\rho$, so the corresponding T^{uu} block is a multiple of the identity and has effective source-side rank 1 (it adds the same contrast to all three components). The monopole $M_0 = \Delta\kappa V\,\varepsilon_{kk}$ couples the one-dimensional trace subspace to itself with scalar $\Delta\kappa$, contributing rank 1. The quadrupole $\bar{M}_{ij} = 2\Delta\mu V\,\varepsilon^{\mathrm{dev}}_{ij}$ couples the five-dimensional deviator

subspace to itself with scalar $\Delta\mu$, so $T^{\varepsilon\varepsilon}_{\ell=2}$ is a multiple of the identity on $\mathbb{R}^5$ and contributes rank 5. Total $1 + 1 + 5 = 7$.

(ii). Suppose $V' \subset \mathbb{C}^9$ is a subspace of dimension $d < 9$ that supports all three channels. By SO(3) invariance the projection onto each irreducible sector must be either zero or full-rank on that sector. The monopole sector has dimension 1, the dipole 3, and the quadrupole 5. If all three projections are full-rank, then V' contains the direct sum $1 \oplus 3 \oplus 5 = \mathbb{C}^9$, so $d = 9$, a contradiction. Hence at least one sector is absent, and the corresponding Lamé contrast cannot be represented. □

Remark 9. Theorem 8 is the reason the solver in `GlobalMatrix/block_riccati_cluster.py` uses a 9×9 T-matrix rather than the smaller 3×3 (displacement-only) or 6×6 (strain-only) options: both of the latter two would silently discard one of the three Lamé-contrast channels and yield a quantitatively wrong Foldy-Lax solution. A 4×4 "acoustic" basis carrying $(p, \boldsymbol{v})$ recovers only the monopole and dipole channels and misses the quadrupole entirely.

8 Conclusions

The three effective multipole sources emitted by a small elastic inclusion are in one-to-one correspondence with the three independent Lamé contrasts and with the three SO(3) irreducible components of the nine-dimensional displacement-strain state vector. Specifically, the monopole channel ($\ell = 0$, driven by $\Delta\kappa$) lives in the trace of the strain; the dipole channel ($\ell = 1$, driven by $\Delta\rho$) lives in the displacement; and the quadrupole channel ($\ell = 2$, driven by $\Delta\mu$) lives in the deviator of the strain. The three strain components carrying a normal derivative across a horizontal interface — ε_{zz}, ε_{xz} and ε_{yz} — are the *dependent* components, and they are exactly those on which the monopole and quadrupole channels write their signatures via the closed-form jump relations (4.1)–(4.1). The dipole channel does not appear in the kinematic strain jumps at all but instead acts dynamically via the continuous displacement field. An elementary parity argument, sharpened by Schur's lemma, gives the local T-matrix a block-diagonal structure that reflects this decomposition, and the nine-component basis is minimal in the sense that no smaller subspace can carry all three Lamé-contrast channels simultaneously.

This structural understanding underpins the block-preconditioned Foldy-Lax solver of the companion paper [7]: the 9×9 basis is *necessary* to capture the full physics, the block-diagonal T-matrix structure justifies caching the local scattering operator by contrast channel, and the parity-based selection rules can be exploited in a future optimisation to reduce the arithmetic of the 2D FFT convolution from nine to the three independent sector sizes 1, 3 and 5.

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